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SMART IRONING DEVICE
To Protect Your Clothes

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During ironing of clothes, they can get burnt, if the iron is kept over a cloth unattended. This usually happens when you get a phone call while ironing or get distracted by some other activity. To overcome this issue, this proposed solution automatically controls the iron wirelessly through a detachable motion sensor.

During ironing, the accelerometer and gyroscope sensor used in the project detects the motion and position (vertical or horizontal) of the iron and sends the information to the wireless control unit. The iron is switched on only when it is in motion and is in its normal horizontal position for ironing. Otherwise, it is kept switched off through the wireless control unit.

The project has a transmitting and a receiving section. The bill of material for the transmitter and receiver sections are shown in Table 1 and Table 2, respectively. Fig. 1 shows the author's prototype with the transmitting device on the left and the receiving device on the right.

Fig. 1: Author's prototype with the transmitting device (left) and the receiving device

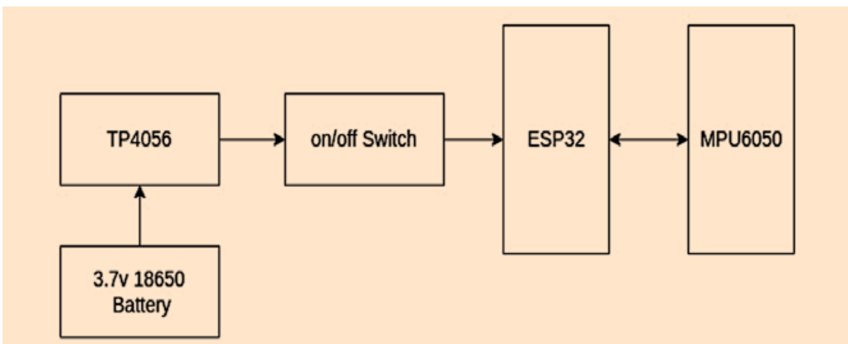

Fig. 2: Block diagram of the transmitter

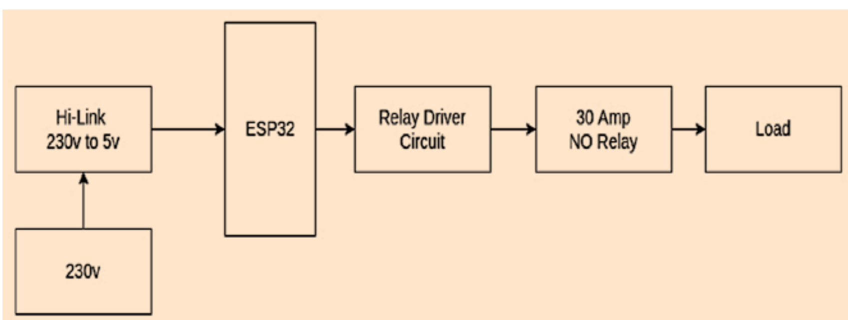

Fig. 3: Block diagram of the receiver

Circuit and working

This project has a sensing part and a triggering part. The sensing part is the transmitter and the triggering part is the receiver. Fig. 2 and Fig. 3 show block diagrams of the transmitter and receiver, respectively.

The transmitter uses MPU6050 as the sensor to detect the iron's movement; it can detect movement in x-axis, y-axis, as well as z-axis. This sensor is I2C compatible; the data it acquires is processed by the ESP32 microcontroller. The ESP32